

A Multi-Lingual Cyberbullying Detection and Blockchain-Integrated Privacy-Preserving Federated Learning Framework

Foyisal Ahmad¹, Sourave Podder¹, Shoumik Karmokar¹, Md. Sanowar Hossain¹, Md. Mahbub Or Rashid^{1,2}, Md. Shakil Ahmed^{1,2,*}, and Sumaiya Binte Shahid^{1,2}

¹ Department of Computer Science and Engineering (CSE), Bangladesh University of Business and Technology (BUBT), Dhaka, Bangladesh

² Institute of Information Technology, Jahangirnagar University, Savar, Dhaka 1342, Bangladesh

Abstract. With the prevalence of social media and the development of multilingualism, the severity and impact of cyberbullying, as well as its psychological and social implications, have exacerbated the issue. In addition to that, the majority of current cyberbullying detection methods are based on centralized data collection and therefore are vulnerable to privacy leakages, illegal data access, and security loopholes. This paper presents a privacy-preserved and multilingual cyberbullying detection framework which leverages Natural Language Processing (NLP), hybrid BiLSTM-LSTM and hybrid Random Forest-XGBoost deep learning techniques, Federated Learning (FL), Blockchain and Explainable Artificial Intelligence (XAI) approaches to alleviate these issues. Our framework takes into consideration English, Bangla, and Banglish social media posts and preprocesses them through text normalization, emoji extraction, PII masking, tokenization and sequence padding. FL facilitates collaborative training without exposing user-level raw data, and the Blockchain secures information by utilizing SHA-256 hashing, Hash-based Message Authentication Code (HMAC), Merkle Root validation and Proof-of-Work (PoW) consensus. Additionally, the application of SHAP and LIME ensure both global and local model interpretability. Experimental results show that our hybrid BiLSTM-LSTM approach is effective for classification accuracy with 98.87%, significantly outperform hybrid Random Forest-XGBoost approach with 93.45%.

Keywords: Cyberbullying Detection · Privacy-Preserving Federated Learning · Blockchain · Hybrid Deep Learning · Hybrid Machine Learning · Multilingual Natural Language Processing · Explainable Artificial Intelligence · SHAP · LIME

1 Introduction

In today's digital era, social media has become a major platform for communication and information sharing. However, its widespread use has also increased

cyberbullying, which involves abusive or harmful online behavior and can cause serious psychological effects, including anxiety, depression, and low self-esteem. The growing use of multilingual and code-mixed content further increases the complexity of automatic cyberbullying detection. Recent advances in Artificial Intelligence (AI) and Natural Language Processing (NLP) have significantly improved cyberbullying detection. Transformer-based models such as BERT and MuRIL, together with traditional machine learning methods, have shown strong performance in detecting offensive content across multiple languages [1, 2]. In addition, explainable AI techniques have enhanced model transparency and reliability [3, 4]. Nevertheless, studies on low-resource languages such as Bangla and Banglish highlight the need for language-specific models and high-quality multilingual datasets [5–10]. Despite these advances, existing cyberbullying detection systems often rely on centralized data collection, increasing privacy risks. Moreover, multilingual and code-mixed text remains challenging for accurate classification. Federated Learning (FL) addresses these issues by enabling collaborative model training without sharing raw user data, although it introduces challenges related to secure aggregation and transparency. Blockchain technology complements FL by providing tamper-resistant, transparent, and decentralized model management. Therefore, this study proposes a Multi-Lingual Cyberbullying Detection and Blockchain-Integrated Privacy-Preserving Federated Learning Framework that integrates NLP, hybrid learning models, FL, and Blockchain for secure and effective multilingual cyberbullying detection across Bangla, English, and Banglish social media text.

The main contributions of this research are summarized as follows:

- A privacy-preserving multilingual cyberbullying detection framework integrating Federated Learning (FL) and Blockchain for secure detection without sharing raw user data.
- A hybrid learning approach combining Hybrid BiLSTM–LSTM and Hybrid RF–XGBoost models to improve multilingual cyberbullying detection accuracy.
- A comprehensive NLP preprocessing pipeline with SHAP and LIME-based explainability to enhance data quality, model transparency, and interpretability.

2 Literature Review

The ever growing multilingual and code-mixed interactions on social media has given rise to the important research field of cyberbullying detection. Researchers have been exploring methods such as machine learning (ML), deep learning (DL), transformer based networks, privacy preserving methods to tackle the issue of cyberbullying detection in low resource, multilingual environment and to address issues such as code-mixing in social media text. Patel *et al.* [11] tried to investigate the issues of code-mixed discourse in social media and Verma *et al.* [12] came up with a transformer-based approach for resource-constrained languages and achieved better accuracy in classification. Some influential works have

been done in the field for Bengali and multilingual contexts; Rahman *et al.* [13] created a big corpus for Bengali cyberbullying and Shah *et al.* [14] developed a multilingual cyberbullying classifier. Gupta *et al.* [15] developed a robust Multilingual Cyberbullying Discovery Framework (MCDF). Yadav and Mishra [16] showed the ML based approaches to detect and prevent cyber bullying. The DL models became equally popular for cyberbullying detection. Roy *et al.* [17] proposed a hybrid of LSTM and CNN model for detection of cyberbullying. Hoque *et al.* [18] demonstrated the problem of multilingual cyberbullying detection in low resource settings. Ranasinghe *et al.* [19] investigated the use of multilingual BERT to detect offensive content while Ranasinghe and Zampieri [20] designed a text-to-text, multilingual offensive language detection model. Mahmud *et al.* [21] also compared ML and DL techniques for cyberbullying detection. Sharma *et al.* [22] proposed a privacy preserving federated learning (FL) framework for cyberbullying detection. Jahin *et al.* [23] improved the Bengali cyberbullying detection using DL approaches. Enhanced detection performance was seen by Alqahtani and Ilyas [24] using ensemble methods and Khafajeh *et al.* [25] using DL techniques. Recent advancements utilized explainable AI, multimodal learning, transformer fusion, and federated privacy-preserving networks. Hasan *et al.* [26] developed BanCyB, Alqaraleh *et al.* [27] proposed a lightweight transformer architecture, Premkumar and Nachimuthu [28] designed a federated graph transformer-based system, and Cao *et al.* [29] and Ismail *et al.* [30] implemented feature-fusion and multimodal models for cyberbullying detection, respectively. Open research problems like privacy, decentralized learning, imbalance of multilingual data, explainability, and secure sharing of models still remain. In this work, we will develop a hybrid, FL, Blockchain and explainable AI based multilingual cyberbullying detection system that provides efficient, privacy-preserving, accurate and scalable detection.

3 Proposed System Architecture

The proposed framework integrating Federated Learning (FL), Blockchain, hybrid learning models, and Explainable AI (XAI) for privacy-preserving multilingual cyberbullying detection as shown in Figure 1. The data are preprocessed and distributed across FL clients for local model training, while the central server aggregates the updates using FedAvg. Blockchain ensures secure aggregation, and SHAP and LIME provide model interpretability. The global model is evaluated using 5-fold CV and standard performance metrics.

Algorithm 1 Blockchain-Integrated Privacy-Preserving Federated Learning

Input: Dataset D , clients N , rounds R

Output: Global model M_G , prediction $\hat{y}$, ledger $\mathcal{B}$

- 1: Preprocess D , split into $(D_{tr}, D_{val}, D_{te})$, and distribute D_{tr} among N clients
 - 2: Initialize global model M_G
 - 3: **for** $r = 1$ to R **do**
 - 4: **for** $i = 1$ to N **do**
 - 5: Train models locally on D_i and obtain update w_i
 - 6: Transmit model updates and metadata; record update integrity in $\mathcal{B}$
 - 7: **end for**
 - 8: $w_G \leftarrow \sum_{i=1}^N \frac{n_i}{n} w_i$
 - 9: $M_G \leftarrow \text{UPDATE}(w_G)$
 - 10: **end for**
 - 11: $\hat{y} \leftarrow M_G(D_{te})$
 - 12: Evaluate using 5-fold stratified CV and explain predictions with SHAP/LIME **re-**
turn $M_G, \hat{y}, \mathcal{B}$
-

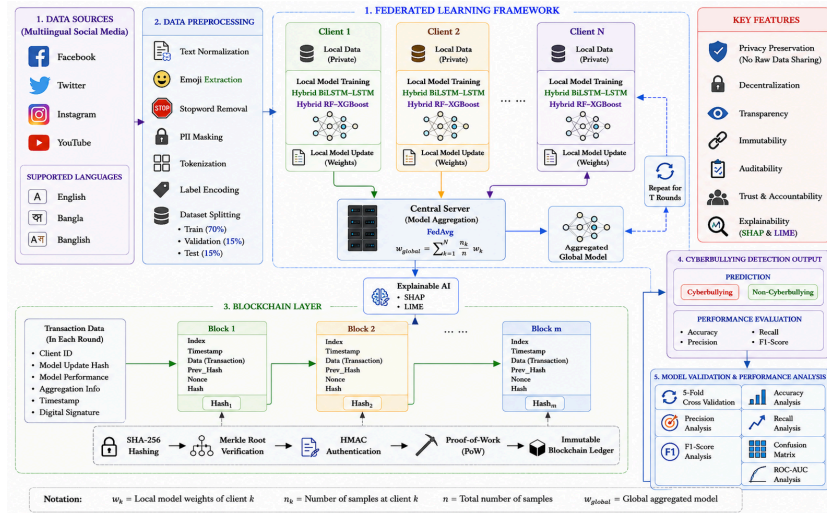


Figure 1. System Architecture for Multilingual Cyberbullying Detection

3.1 Data Collection and Preprocessing

Multilingual data collection comprised 21,204 social media posts in Bangla, English, and Banglish from Facebook, Instagram, Twitter, and YouTube (2023–2025), including 12,384 (58%) cyberbullying and 8,820 (42%) non-cyberbullying posts. Language-wise, the dataset contained 5,083 Bangla, 9,703 Banglish, and 6,418 English posts. Duplicates and invalid entries were removed, followed by a leakage audit that eliminated 146 overlapping training instances and confirmed zero exact cleaned-text overlap between training and testing data. URL/special-

character removal, normalization, stop-word processing, feature encoding, and PII masking were then applied.

Table 1: Dataset Attribute Overview

Attribute	Type	Description	Values
ID	Integer	Unique record identifier.	1-21204
Text	String	Social media post/comment.	Free text
Language	Nominal	Language of the text.	English, Bangla, Banglish
Platform	Nominal	Source platform.	Facebook, X, YouTube, Instagram
Label	Binary	Target class.	Cyberbullying, Non-Cyberbullying
Type	Nominal	Cyberbullying category.	Harassment, Hate Speech, Toxic, Threat
Gender	Nominal	User gender (if available).	Male, Female, Other
Year	Integer	Collection/posting year.	2023-2025
Age	Integer	User age (if available).	Dataset-specific

3.2 System Architecture

The framework proposed above is demonstrated in Figure 2, where preprocessed multilingual social media data are processed through Blockchain-based Federated Learning approach. Model explainability is provided by SHAP and LIME methods, while performance is assessed through standard metrics.

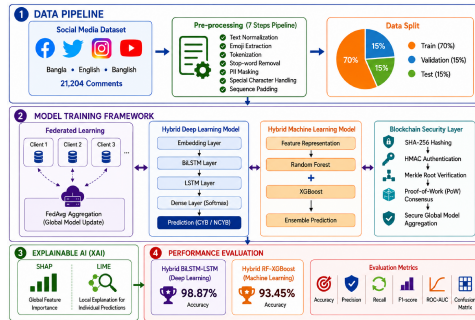


Figure 2. Proposed System Architecture for Multilingual Cyberbullying Detection Framework

3.3 Federated Learning Architecture

The proposed Federated Learning (FL) scheme for privacy-preserving multilingual cyberbullying detection is illustrated in Figure 3. Raw multilingual text remains locally at the clients, where the Hybrid BiLSTM-LSTM model is trained, while only model parameters are transmitted to the central server for Federated Averaging (FedAvg) [31]:

$$w_{\text{global}} = \sum_{k=1}^N \frac{n_k}{n} w_k \quad (1)$$

where w_{global} denotes the aggregated parameters, w_k the parameters from client k , n_k its local training samples, and n the total samples across clients. The FL process includes global-model distribution, local training, update transmission, and FedAvg aggregation over four communication rounds across three participating clients. Model updates are logged through Blockchain to support integrity, accountability, traceability, and tamper-resistant auditing.

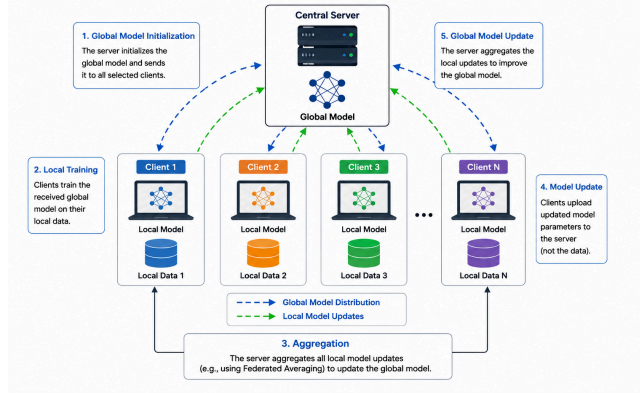


Figure 3. Proposed Federated Learning Framework

3.4 Blockchain Integration

Figure 4 illustrates the Blockchain-integrated Federated Learning framework. Local clients train the Hybrid BiLSTM-LSTM model, while only model parameters and metadata are shared with the server. SHA-256, HMAC, Merkle Root validation, and lightweight Proof-of-Work (PoW) support model-update integrity, authenticity, and tamper-resistant auditing. The threat model considers unauthorized modification of model updates during communication, with the blockchain mechanisms enabling tampering detection without exposing raw training data. Comprehensive poisoning, inference-attack, and blockchain-overhead evaluations are positioned as next-stage empirical validation extensions.

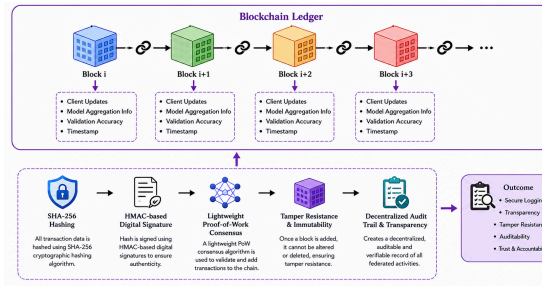


Figure 4. Proposed Blockchain Integration Framework

3.5 Hybrid Deep Learning Model

Hybrid BiLSTM–LSTM Model The new presented Hybrid BiLSTM–LSTM model enhances the task of multilingual cyberbullying detection using the Blockchain-integrated FL model. This hybrid model comprises an embedding layer, BiLSTM and LSTM layers as context and sequence learning layers to represent sentence features, then applies dropout and sigmoid classifier for binary classification. Training of the model on the distributed clients, without any raw data sharing ensures the safe detection of cyberbullying, hate speech, abusive texts in Bangla, Banglish and English language from the social media.

3.6 Hybrid Machine Learning Models

Hybrid Random Forest–XGBoost Model The proposed Hybrid RF–XGBoost model improves multilingual cyberbullying detection within the Blockchain-integrated Federated Learning (FL) framework. RF captures complex patterns, while XGBoost enhances classification through gradient boosting. The model is trained locally on distributed clients, where only model parameters are shared to preserve privacy. Its mathematical formulation is given by [32]

$$\hat{y} = f(RF(x), XGB(x)) \quad (2)$$

where $RF(x)$ and $XGB(x)$ denote the predictions of the Random Forest and XGBoost classifiers, respectively, $f(\cdot)$ represents the ensemble decision function, and $\hat{y}$ is the final predicted class.

3.7 Explainable Artificial Intelligence (XAI)

SHAP interoperability SHapley Additive exPlanations (SHAP) was used to provide global interpretability of the proposed framework by quantifying the contribution of each input feature to the prediction. This improves the transparency and interpretability of multilingual cyberbullying detection. The Shapley value is computed as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (3)$$

where ϕ_i denotes the contribution of feature i , N is the set of all input features, and $f(\cdot)$ represents the prediction function.

Lime interoperability Local Interpretable Model-agnostic Explanations (LIME) was used to provide local interpretability by explaining individual predictions of the proposed hybrid models. It approximates the original model with an interpretable surrogate model, enabling feature-level explanations for each prediction.

This enhances the transparency and trustworthiness of the proposed multilingual cyberbullying detection framework. The LIME surrogate model is defined as:

$$\xi(x) = \arg \min_{g \in G} L(f, g, \pi_x) + \Omega(g) \quad (4)$$

where f is the original prediction model, g is the interpretable surrogate model, π_x defines the locality around the instance x , L measures the approximation error, and $\Omega(g)$ represents the complexity of the surrogate model.

4 Results and Discussion

This section presents a comprehensive performance evaluation of FL models, CV models, and the proposed model.

4.1 Performance Across Federated Learning Model

Table 2: Validation Accuracy of the Hybrid BiLSTM–LSTM Model Across Federated Learning Rounds

Round	Hybrid BiLSTM–LSTM Accuracy (%)
1	59.46
2	78.14
3	94.08
4	98.46

Hybrid BiLSTM–LSTM validation accuracy across four FL rounds is reported in Table 2, increasing from 59.46% to 98.46%. FedAvg enables efficient federated training by sharing only model parameters. The model size is 15.4130 MB, with 369.9129 MB total communication overhead across four rounds. A label-flipping attack was applied to Client 1 to assess FL robustness.

4.2 Blockchain Verification and Overhead Results

The Blockchain layer recorded 22 blocks, 0.010921 MB storage, and 0.000007 s verification time, with 10.3195 ms/sample inference time.

4.3 Cross-Validation Performance Evaluation

We summarize model performance using 5-fold CV, as reported in Table 3. The Hybrid BiLSTM–LSTM achieved 0.99888 accuracy and 0.998856 F1-score with a standard deviation of 0.000994, indicating stable performance while highlighting the need for further validation of generalizability.

Table 3: Performance evaluation of the proposed hybrid models using 5-fold cross-validation.

Model	Fold	Precision	Recall	F1-Score	Accuracy
Hybrid BiLSTM–LSTM	F1	0.999	0.999	0.999	0.999
	F2	0.999	0.999	0.999	0.999
	F3	1.000	0.999	0.999	0.999
	F4	0.999	1.000	0.999	0.999
	F5	0.997	0.997	0.997	0.997
	Mean	0.999	0.999	0.999	0.999
	SD	0.001	0.001	0.001	0.001
Hybrid RF–XGBoost	F1	0.998	0.998	0.998	0.998
	F2	0.998	0.998	0.998	0.998
	F3	0.999	0.999	0.999	0.999
	F4	0.999	0.999	0.999	0.999
	F5	0.997	0.997	0.997	0.997
	Mean	0.998	0.998	0.998	0.998
	SD	0.001	0.001	0.001	0.001

4.4 Confusion Matrix

Confusion matrix measures classification based on four different parameters: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Increased TP and TN values mean better classification results.

Table 4: Classification Prediction Summary of the Hybrid Models

Model	Precision	Recall	F1-Score	TP	FP	FN	TN
Hybrid BiLSTM–LSTM	0.9905	0.9864	0.9883	1287	0	36	1852
Hybrid RF–XGBoost	0.9493	0.9215	0.9308	1116	1	207	1851

Table 4 summarizes and analyzes the confusion matrix and classification results of our proposed hybrid models. The Hybrid BiLSTM–LSTM had the best performance, achieving a 0.9905 Precision, 0.9864 Recall, and 0.9883 F1-score, which performed better than Hybrid RF–XGBoost. The results illustrate the capabilities of Hybrid BiLSTM–LSTM on detecting multilingual cyberbullying.

4.5 Performance Analysis of Proposed Models

Table 5 compares the proposed models. The Hybrid BiLSTM–LSTM model achieved 98.75% training and 98.87% testing accuracy, outperforming Hybrid RF–XGBoost. These results indicate strong performance in multilingual cyberbullying detection; however, broader validation is needed to further assess generalizability.

Table 5: Training and Testing Performance Comparison of the Proposed Hybrid Models

Hybrid Model	Training Performance (%)				Testing Performance (%)			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Hybrid BiLSTM-LSTM	98.75	98.94	98.51	98.71	98.87	99.05	98.64	98.83
Hybrid RF-XGBoost	92.65	94.38	91.19	92.20	93.45	94.93	92.15	93.08

Table 6: Test Accuracy Comparison of the Proposed Hybrid Models

Hybrid Model	Test Accuracy (%)
Hybrid BiLSTM-LSTM	98.87
Hybrid RF-XGBoost	93.45

As presented in Table 6, Hybrid BiLSTM-LSTM has obtained the highest test accuracy which is 98.87% among Hybrid RF-XGBoost model indicating its efficacy for the task of multilingual cyberbullying detection.

4.6 Explainable AI Analysis

Figure 5 presents the SHAP-based interpretability analysis of the proposed hybrid models. Figures 5(a) and (b) show the SHAP summary and waterfall plots of the Hybrid BiLSTM-LSTM model, providing global feature importance and local prediction explanations. Similarly, Figures 6(c) and (d) present the SHAP summary and waterfall plots of the Hybrid RF-XGBoost model. Positive SHAP values indicate features that increase the probability of cyberbullying prediction, while negative values reduce the prediction contribution. The results demonstrate that both hybrid models capture important multilingual patterns and improve the interpretability and reliability of the proposed framework.

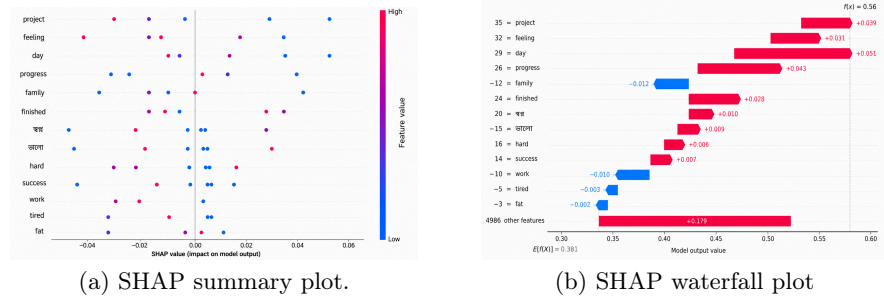


Figure 5: SHAP visualizations of the Hybrid BiLSTM-LSTM model.

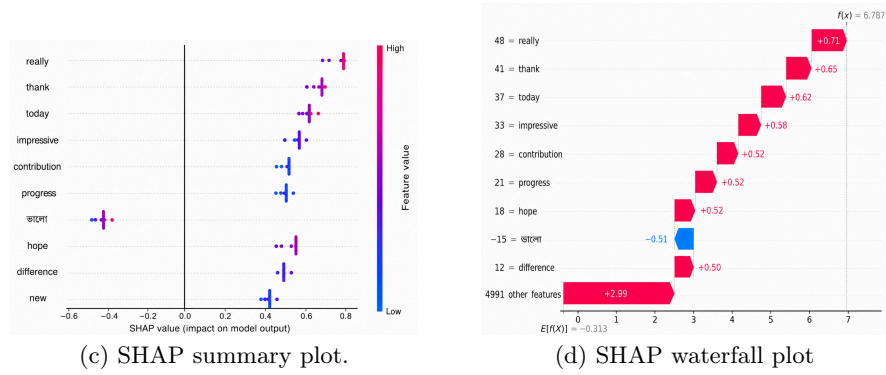


Figure 6: SHAP visualizations of the Hybrid RF-XGBoost model.

In Figure 7, a LIME explanation of one example prediction is shown. Highlighted words indicate influential features and predicted probabilities show model's confidence in this case. Thus it suggest that local interpretability of model should depend on meaningful patterns within text.

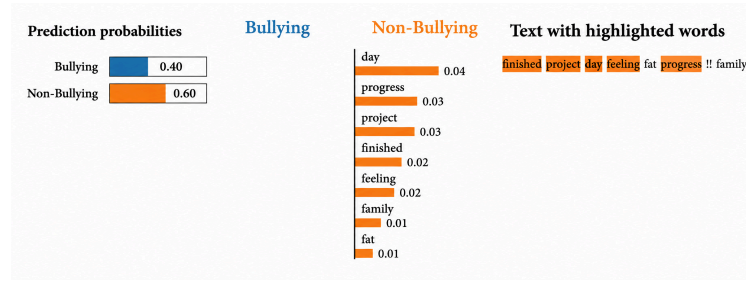


Figure 7. LIME explanation highlighting influential words for the predicted class.

4.7 ROC Curve Analysis

The ROC curves of the proposed hybrid models were displayed in Figure 8. Comparing both hybrid models, Hybrid BiLSTM-LSTM model was ranked higher in AUC, with an accuracy of 0.9991 than Hybrid RF-XGBoost model of 0.9967. All the proposed hybrid models performed very well with high discriminative power, and Hybrid BiLSTM-LSTM was the best model for multilingual cyberbullying detection.

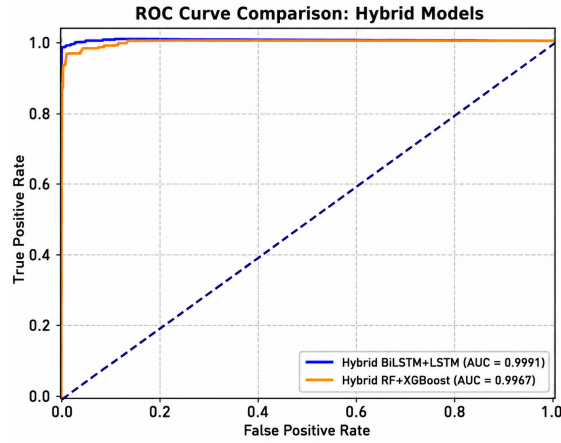


Figure 8. ROC curve of the proposed hybrid model.

4.8 Comparison with Previous Papers

Table 7: Comparison of the Proposed Framework with Existing Cyberbullying Detection Approaches

Reference	Model	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
S. K. Pradhan et al. [33]	Blockchain + FL + Transformer	Multilingual Social Media	96.8%	96.6%	96.2%	96.4%
A. N. Khan et al. [34]	Federated Transformer + BERT	Code-mixed + English Social Media	95.9%	95.7%	95.4%	95.6%
L. Zhang et al. [35]	Secure FL (CNN + LSTM + Blockchain)	Multilingual Abuse Dataset	97.1%	97.0%	96.8%	96.9%
P. Saha et al. [36]	FL + Differential Privacy + XLM-R	Cross-lingual Social Media Dataset	96.3%	96.1%	95.9%	96.0%
M. A. Hossain et al. [37]	Multimodal FL + Blockchain Aggregation	Low-resource Multilingual Dataset	95.7%	95.5%	95.2%	95.4%
Proposed Framework	Hybrid BiLSTM-LSTM + Hybrid RF-XGBoost	Bangla + English + Banglish	98.87%	99.05%	98.64%	98.83%

Table 7 compares the proposed framework with existing ML, DL, FL, and Blockchain-based privacy-preserving techniques for multilingual cyberbullying detection. To the best of our knowledge, this study is among the few to combine Hybrid BiLSTM-LSTM and Hybrid RF-XGBoost within an FL-Blockchain architecture for Bangla, English, and Banglish cyberbullying detection. The proposed framework achieved strong performance, with 98.87% accuracy, 99.05% precision, 98.64% recall, and 98.83% F1-score.

4.9 Discussion

Experimental results demonstrate the effectiveness of FL and Blockchain. FL enables training without raw data sharing, while Blockchain supports model-update integrity and auditing. Hybrid BiLSTM-LSTM outperformed Hybrid

RF-XGBoost, with SHAP and LIME improving interpretability. Future work will address generalization, adversarial evaluation, and system overhead.

5 Conclusion

This paper presents a framework for multilingual cyberbullying detection by integrating Blockchain technology with Federated Learning. The experimental results demonstrate effective cyberbullying detection while keeping raw training data at the client side and securely auditing model updates. The Hybrid BiLSTM-LSTM model achieved strong classification performance, while SHAP and LIME enhanced prediction interpretability.

Code and Data Availability

The corresponding author can provide the code and data upon reasonable request at [shakil.ahmed@bubt.edu.bd].

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